MB Note

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K})$ N: population at time t; K: carrying capacity (depends on resources); r,K: >0.

Sol: $N(t) = \frac{N_0 K e^{rt}}{K+N_0(e^{rt}-1)} \xrightarrow{\infty} K$ **Equilibrium points:** $N=0$ (unstable), $N=K$ (stable). **Quickest:** $N=\frac{K}{2}$ (<: 凹; >: 凸)

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t-T))$ T: delay time >0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

[1]**Ext. of Logistic Model:** $\frac{dN}{dt} = rN(t)\left[1-\frac{N(t-T)}{K}\right]$ (用"ave. pop." 估计, 准用"Convolution Type") $\Rightarrow$ [2]**Dimensionless:** $\frac{dN^*}{dt^*} = N^*(t^*)\left[1-N^*(t^*-T^*)\right]$ ($N^*=\frac{N}{K}, t^*=rt, T^*=rT$)

[1,2] **Periodic sol:** stable limit cycle $N(t+t_p)=N(t), t_p$: period. **Amplitude:** rT **Equilibrium Points:** $N=0|N^*=0$ (unstable); $N=K|N^*=1$ (Stable: $0<T^*=rT<\frac{\pi}{2}$; Unstable: $T^*>\frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K}) - EN = f(N)$ r,K,E: >0; EN: harvesting rate/unit time; E: effort measure.

Equilibrium: $N=0$ (unstable); $N=N_h(E)=K(1-\frac{E}{r})>0$ (need $E<r$, Stable) **Yield:** $Y(E)=EN$ $\cdot E>r$:die out; else: **Max Yield:** $Y_M=\frac{rK}{4}$ $(E=\frac{r}{2})$ **Steady State:** $N_{h|Y_M}=\frac{K}{2}$

Continuous Population Model (Y_0): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K}) - Y_0 = f(N;r,K,Y_0)$ Y_0: constant yield rate/unit time.

Max Yield: $Y_M=\frac{rK}{4}$ **Equilibrium:** [1]$0<Y_0<Y_M$: Two positive points (unstable(小) and stable(大)). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1}=f(N_t)$: Verhulst: $N_{t+1}=rN_t(1-\frac{N_t}{K}); r,K>0$. Better: **Ricker curve:** $N_{t+1}=N_t e^{r(1-\frac{N_t}{K})}$

Imperfect Bifuraction: $\dot{x}=rx-x^3+h$ (codimension-2 bifurcation) If $r\le 0$, x^* unique. If $r>0, |h|>h_c, x^*$ unique; $|h|<h_c, x^*$ three. (两边 stable, 中 unstable). Bifuraction points: $h=\pm h_c=\pm\frac{2r}{3}\sqrt{\frac{r}{3}}$. **cusp point:** $(r,h)=(0,0)$

1.2 Summary - Stability and Bifuraction

	$\frac{dN}{dt}=f(N)$ **No Limit cycle periodic solutions** S: stable; U:unstable.				$N_{t+1}=N_tF(N_t)=f(N_t)$ **Eigenvalue:** $\lambda=f'(N^*)$			
Stability	Stable: $f'(N^*)<0$		Unstable: $f'(N^*)>0$		Stable: $\|\lambda\|<1$	Unstable: $\|\lambda\|>1$ (chaotic)	Periodic: $\|\lambda\|=1$ $\begin{cases}+: \text{tangent bifurcation}\\ -: \text{period-doubling bifurcation}\end{cases}$ (neutrally stab.)	
Linearise	$N(t)=N^*+n(t), \|n(t)\|\ll 1 \Rightarrow \frac{dn}{dt}=\frac{dN}{dt}=\begin{cases}\text{(general)}=f(N^*)+f'(N^*)n(t)\\ \text{(delay)= 直接带入 f 后近似}\end{cases}$				Let $N_t=N^*+n_t, \|n_t\|\ll 1 \Rightarrow N^*+n_{t+1}=\cdots=f(N^*+n_t)\approx f(N^*)+f'(N^*)n_t \Rightarrow n_{t+1}=\lambda n_t$			
Bifuraction \\ Perturbations at N^* | Periodic	Saddle-Note (Normal Form) $\dot{x}=r\pm x^2$ 0→S→U-S	Transcritical $\dot{x}=rx-x^2$ S-U→U-S	(super-)Pitchfork $\dot{x}=rx-x^3$ S→S-U-S	Sub-critical Pitchfork $\dot{x}=rx+x^3$ U→U-S-U	$\lambda<-1$: ↑ oscillations	$-1<\lambda<0$: ↓ oscillations	$0<\lambda<1$: ↓ Monotone	$\lambda>1$: ↑ Monotone
					If $f^p(N^*)=N^*$, N^* is p-periodic. | $\forall$ even p-periodic N^* with $\lambda=-1$ branches into $2p$-periodic			
					Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p\ge 1$, p-periodic N^* exists.			**Species extinction:** $N_{\min}<1$
Other	**Time-Independent**: At N^*: $\frac{dN}{dt}=0$ $N(t)=N(t-T)=N^*$ $f(N^*)=0$				**Max/Min:** [如果单峰 (unimodal)|2-periodic] $f'(N_m)=0, N_{\max}=f(N_m)$; $N_{\min}=f(N_{\max})$ $\downarrow E, Y_0$ 对应两种模型			
Other	**Tangency Condition**: If $\dot{x}=f(x,r)$, 分岔点 (x^*,r_c): $\frac{\partial f}{\partial x}\|_{(x^*,r_c)}=0$ $f(x^*,r_c)=0$ | 两点合并:**coalesce**.				|| **Recovery Time:** $T=-\frac{1}{\lambda}$ 比收获模型好 看 recovery time → 算 $\frac{T(E)}{T(0)}\|\frac{T(Y)}{T(0)}$ 小好			

Hopf Bifurcation: 平衡点从稳定变为不稳定，同时产生 limit cycle. 条件: $\det(J)>0$; $\frac{d\mathrm{tr}(J)}{d\mu}\ne 0$ or: $\mathrm{tr}(J)$ +/-符号有变化 $\Leftrightarrow$ $\lambda=\pm iw$ and $\frac{d\Re(\lambda)}{d\mu}\ne 0$ μ: parameter, not time.

$\Rightarrow$ Stable → Unstable + Stable Limit Cycle (supercritical) or Unstable → Stable + Unstable Limit Cycle (subcritical). (一般 super-) when $\mathrm{tr}(J)=0|\Re(\lambda)=0, \lambda=\pm iw \Rightarrow$ **periodic:** $T=\frac{2\pi}{w}$

Bifuraction: Qualitative changes in dynamics are called *bifurcations*. (当平衡点性质 | 个数发生变化时) **Bifuraction Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 |For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt}$ $\Rightarrow$ Assume $n(t)=e^{\lambda t}|ce^{\lambda t}$ → Find $\lambda=g(\lambda)$ $\Rightarrow$ Let $\lambda=u+iw \to u=g_1(u)$ and $w=g_2(w)$ $\Rightarrow$ $u<0$ asy.stable; $u>0$ unstable. **Find T_c:** (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T=0$ stable. $\Rightarrow$ Using $u=0$ (i.e. $\lambda=iw$) to find T_c.

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt}=N(a-bP)$; $\frac{dP}{dt}=P(cN-d)$ $\Rightarrow$ nondim: $\frac{du}{d\tau}=u(1-v)$; $\frac{dv}{d\tau}=\alpha v(u-1)$ **Equilibrium:** $(u^*,v^*)=(0,0),(1,1)$

Exact sol: $\alpha u+v-\ln(u^\alpha v)=H, H>H_{min}=\alpha+1$ constant. (closed phase) **Conservation Law:** Using $\frac{du}{dv}=\frac{du}{d\tau}\frac{d\tau}{dv}$ find H $\Rightarrow$ $\frac{dH}{d\tau}=0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u,v)$ is level curve, Look $\begin{bmatrix}H_{uu} & H_{uv}\\ H_{vu} & H_{vv}\end{bmatrix}_{(u^*,v^*)}$ $\Rightarrow$ If $+/-$ definite (i.e. $\forall\lambda_i>0(<0)$)

$\Rightarrow$ It's local min/max at (u^*,v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix}u'(t)\\ v'(t)\end{bmatrix}=\begin{bmatrix}f(u,v)\\ g(u,v)\end{bmatrix}$ at equilibrium (u^*,v^*). Let $\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}u-u^*\\ v-v^*\end{bmatrix}$, then $\begin{bmatrix}x'\\ y'\end{bmatrix}=\begin{bmatrix}f_u & f_v\\ g_u & g_v\end{bmatrix}_{(u^*,v^*)}\begin{bmatrix}x\\ y\end{bmatrix}$ $\Rightarrow$ Find eigenvalues λ of J by $\det(J-\lambda I)=0$

[1] $\det(J)>0, \mathrm{tr}(J)<0 \Leftrightarrow \Re(\lambda)<0 \to$ stable node $_{Im(\lambda)=0}$; stable focus=spiral point $_{Im(\lambda)\ne 0}$; [2] $\det(J)>0, \mathrm{tr}(J)>0 \Leftrightarrow \Re(\lambda)>0 \to$ unstable node $_{Im(\lambda)=0}$; unstable focus=spiral point $_{Im(\lambda)\ne 0}$

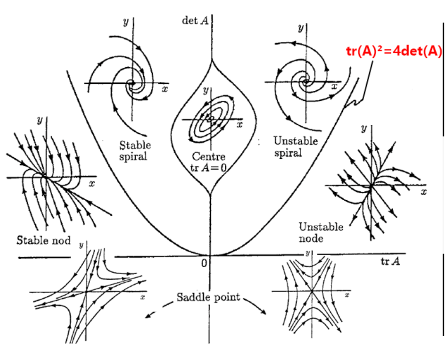
[3] $\det(J)<0 \Leftrightarrow Im(\lambda)=0, \lambda_1<0<\lambda_2 \to$ saddle point (unstable); [4] $\det(J)>0, \mathrm{tr}(J)=0 \Leftrightarrow \lambda=\pm iw$ (pure imaginary) → center (only linear stability)|Hopf bifurcation (见前面)

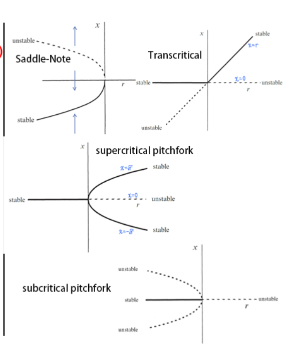
Nullclines: $f(u,v)=0$ or $g(u,v)=0$ → 得 u,v 表达线条是 nullclines. **Trapping Region:** A closed region R 可以用直线/null clines 构成 s.t. 所有 边界的外法线方向 **n**, 在对应边界上的点, 满足 $\mathbf{n}\cdot\begin{bmatrix}f(u,v)\\ g(u,v)\end{bmatrix}<0$

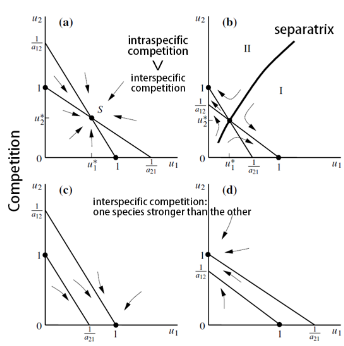
Find slope of Nullclines: $f(u,v)=0 \Rightarrow df=f_u du+f_v dv=0 \Rightarrow \frac{dv}{du}=-\frac{f_u}{f_v}$ Similarly: $\frac{dv}{du}=-\frac{g_u}{g_v}$ ↑ 也叫 bounding region (confined region)

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in *trapping region* $\Rightarrow$ A **stable limit cycle** exists. (Cannot be a saddle (since $\det(J)$ cannot be <0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (b,c,d; b 物种谁最后存活取决于初始值)







Mutualism|Symbiosis: (共生)

Case I: 图 (a) coexist: competition not aggressive.

Case II: 两条射线 both species grow unboundedly.(all unstable)

Case II: 两条向外的线相交在一点 both species tend to a stable coexistence.

For small perturbations: If fixed point is Center or Improper Node (GM=1): Any small perturbation $\Rightarrow$ may change to spiral point. $\Rightarrow$ Not structurally stable.

Chaotic: Three dim or delay $\Rightarrow$ 可能 chaotic.

Limit Cycle: Any sufficiently perturbation from a stable limit cycle $\Rightarrow$ trajectory returns to the limit cycle. (i.e. tends to zero asymptotically with time)

3 Reaction Kinetics

Enzyme Kinetics: e.g. $S+E \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} SE \xrightarrow{k_2} P+E$ K_i: rate of reaction; S: substrate; E: enzyme; SE: complex; P: product. **Rate of uptake:** $v=\frac{d[P]}{dt}|_{t=0}$

Law of Mass Action: Rate of reaction is proportional to the product of the concentrations of the reactants. e.g. $\alpha_1 A+\alpha_2 B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \beta_1 C+\beta_2 D \Rightarrow \text{rate}_{\text{forward}}=k_1[A]^{\alpha_1}[B]^{\alpha_2}$; $\text{rate}_{\text{backward}}=k_{-1}[C]^{\beta_1}[D]^{\beta_2}$

Quasi-Steady State Approximation (QSSA): 如果系统里有快和慢变量, 则可以假设快变量达到稳态, 即 $\frac{d(\text{fast})}{dt}=0$. (or $\varepsilon\to 0$; 一般来说假设 complex concentration changes rapidly)

Singular Perturbations Analysis: For $\frac{du}{d\tau}=f(u,v), \varepsilon\frac{dv}{d\tau}=g(u,v), 0<\varepsilon\ll 1 \Rightarrow v$ fast (when $g\not\to 0$), u slow. **Quasi-steady-state approximation:** $\varepsilon\to 0 \Rightarrow g(u,v)=0$

Taylor: $u(\tau,\varepsilon)=\sum_{n=0}^{\infty}\varepsilon^n u_n(\tau)$; $v(\tau,\varepsilon)=\sum_{n=0}^{\infty}\varepsilon^n v_n(\tau)$ **Outer solution:** Taylor 带入原方程组 → 保留 $\mathcal{O}(1)|\varepsilon=0$ Valid for $\tau>0$ (slow time); $v_0(0)$ **unvalid**.→not uniformly valid for $\tau\ge 0$

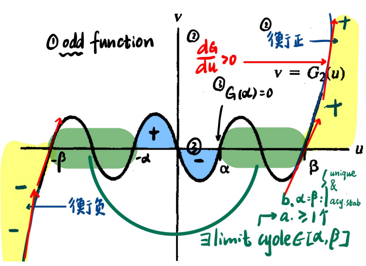
Inner solution: $\sigma=\frac{\tau}{\varepsilon}$, $\substack{U(\sigma;\varepsilon)=u(\tau;\varepsilon)\\ V(\sigma;\varepsilon)=v(\tau;\varepsilon)}$ $\Rightarrow \frac{dU}{d\sigma}=\varepsilon f(u,v)$, $\frac{dV}{d\sigma}=g(u,v)$ → Taylor 带入原方程组 → 保留 $\mathcal{O}(1)|\varepsilon=0$ (known as **boundary layer** ($\tau=0$ 处)); Valid $\tau=0$ (fast time)

Asymptotic Matching: $\lim_{\sigma\to\infty}[U_0(\sigma),V_0(\sigma)]=\lim_{\tau\to 0^+}[u_0(\tau),v_0(\tau)]$ **Uniformly Valid Solution:** $u(\tau;\varepsilon)=u_0(\tau)+\mathcal{O}(\varepsilon)$; $v(\tau;\varepsilon)=\begin{cases}v_0(\tau)+\mathcal{O}(\varepsilon), & 0<\varepsilon\ll\tau\\ V_0(\sigma)+\mathcal{O}(\varepsilon), & 0<\tau\ll 1\end{cases}$ **uniform expansion:** 见 chp.5

Activator-Inhibitor Model: For $\frac{du}{dt}=f(u,v)$, $\frac{dv}{dt}=g(u,v)$: If $\frac{\partial g}{\partial u}>0$, u is an **activator** of v; If $\frac{\partial f}{\partial v}<0$, v is an **inhibitor** of u. ps: Possible have bifurcation behaviour

Positive/Negative Feedback: At Equilibrium: **Positive:** $f_u,f_v\ge 0; g_u,g_v\le 0$ or $f_u,f_v\le 0; g_u,g_v\ge 0$ **Negative:** $f_u,f_v\ge 0; g_u,g_v\ge 0$ or $f_u,f_v\le 0; g_u,g_v\le 0$

4 Relaxation Oscillations & Switch Behaviour

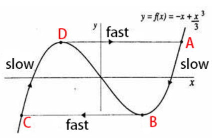


Find Limit Cycle: For $\begin{cases}\dot{u}=v-G(u)\\ \dot{v}=-u\end{cases}$ $\Leftrightarrow$ $\frac{d^2u}{dt^2}+g(u)\frac{du}{dt}+u=0$ with $G(u)=\int_0^u g(s)ds$. Lienard Equations If we have:

1: $G(-u)=-G(u)$ (odd) 2: $\exists\alpha>0$ s.t. $G(\alpha)=0$ and $G(u)<0, \forall u\in(0,\alpha)$ 3: $\exists\beta\ge\alpha$ s.t. $G(u)>0$, $\frac{dG}{du}>0, \forall u>\beta$ and $G(u)\to\infty$ as $u\to\infty$.

Then: G is 'bumpy' function. [1] $\exists$ limit cycle between $(\alpha,\beta),(-\beta,-\alpha)$. [2] unique and asy.stable if $\alpha=\beta$

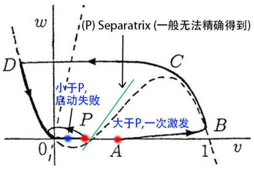
Relaxation Oscillations: A periodic movement of "slow-fast alternation". e.g. For $\begin{cases}\varepsilon\dot{u}=v-G(u)\\ \dot{v}=-u \mid -u\varepsilon\end{cases}$ or More general form: $\begin{cases}\varepsilon\dot{u}=f(u,v)\\ \dot{v}=g(u,v)\end{cases}$ $|\varepsilon|\ll 1$



Relaxation Oscillations requies unstable equilibrium point.

Explain why it occur: [1] Find *nullclines* (一般: 直线 +S 曲线) [2] Since $\dot{u}=\frac{\text{some func}}{\varepsilon}\gg 1$ (so that $|\dot{u}|\gg|\dot{v}|$) unless close to nullcline →(一般 S 曲线) [3] 图上选 4 个点 ABCD (只是一个示例而已), 一般选 S 的 local max/min 加 S 更远处两个点, 连接成一个 loop. [4] 判断 ABCD 点的流向 (看 nullcline 两侧/上下) [5] 分段分析 + 从一不在 nullcline 上的点分析 e.g. $A\to B$ $\dot{u}=0, \dot{v}=-u<0$ 向下, 慢阶段. $B\to C$ not close to S curve, $|\dot{u}|\gg 1$ 向左, 快阶段/fast horizontal jump. $C\to D$ ... $D\to A$ ...

Estimate period T: 例如上图 (例子忽略了快反应) $T = \int_{t_A}^{t_B} dt + \int_{t_C}^{t_D} dt = 2\int_{t_A}^{t_B} dt$ **Find** $dt = \_du$ 带入计算 e.g. In slow time: $\dot{u} = 0 \to v = G(u)$ and $\dot{v} = -u \to \frac{dv}{dt} = \frac{dv}{du}\frac{du}{dt} = G'(u)\frac{du}{dt} = -u \to -\frac{G'(u)}{u}du = dt$, 带入计算



Excitable System: FitzHugh-Nagumo Model $\begin{cases} \frac{dx}{dt} = c[y + x - \frac{x^3}{3} + z(t)] \\ \frac{dy}{dt} = \frac{-x+a+by}{c} \end{cases}$ (x: excitability variable; y: recovery variable; $z(t)$: external stimulus)

Threshold 现象: **Small perturbation**: return to equilibrium (failed initiation/excitation). **Large perturbation**: large excursion before returning to equili. (action potential|successful initiation). After excitation| 大扰动之后: Refractory Period不应期: system is unresponsive to further stimulation. **constant stimulus**: periodic firing (limit cycle).

Reason: 寻找 $\dot{u}$, $\dot{v}$ s.t. $\ll, \gg 1$ i.e. $|\dot{u}| \gg |\dot{v}|$ or $|\dot{u}| \ll |\dot{v}|$

Switch Behaviour: 从一个 stable point 到另一个 stable point. (前面 excitable system|relaxation oscillation 如有两 stable points 且画图分析 (像 relaxation oscillation 那样分析) 可得到)

描述话语 example: 快-慢震荡: [1](option) The equilibrium point S is unstable between the two *knees/turnning points* of the u-nullcline/cubic-like curve. | or: S changes stability as it crosses the knees of the u-nullcline. [2] 图上标/简单解释拐点位置. [3] 证明 $|u'| \gg 1$(写成乘积形式判断)或其他 → 得快运动边. [4] Trajectories *starting off* the u-nullcline *repidly/quickly* approach *left/right branch* of the u-nullcline, and then *slowly move down/up|crawl along* the u-nullcline due to $v' < 0/ > 0$, and to *upper/lower knee/turnning point* [5] At the knee/turnning point, the trajectory *quickly jumps* to the *right/left branch* of the u-nullcline, and then *slowly moves up/down|crawl along* the u-nullcline due to $v' > 0/ < 0$. [6] And this process *repeats periodically*, giving rise to a *relaxation oscillation*. **Excitable**: 类似的, 分析 small perturbation *above/below* equilibrium point S -> 得 *failed excitation* or *successful excitation*.

5 Manifolds & Fenichel's Theorem | SIR Model

Regular/Singular Perturbation: 对于一个含 ε 的方程/系统: If $\varepsilon \to 0$ the order does not change, it's a **regular perturbation**. If $\varepsilon \to 0$ the order changes, it's a **singular perturbation**.

Find Roots: Assume roots $x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \cdots$, 代入比较 $\mathcal{O}(1), \mathcal{O}(\varepsilon)$... 得系数(解前两系数就行). If *singular*, cannot approx $x = \mathcal{O}(\varepsilon^{-1})$, rescaling $x = \frac{X}{\varepsilon}$, assume root $X = X_0 + \varepsilon X_1 + \cdots$ 类似前. 不留 $X_0 = 0$ 解, 因为 X 小.

Sol & Uniform: 对 $\varepsilon\ddot{y} + p(x)\dot{y} + q(x)y = r(x)$, a. 类似上面算 sol (有初值) or b. 直接算 exact sol. → 得**Outer Solution** (首项含 ε). Let $\tau = \varepsilon t$ 类似算 sol → **Inner Solution** (首项不含 ε). **unif expre** = Outer + Inner - Matching(=matching cond 的 lim)

Singular Perturbation System: For $\begin{cases} \dot{u} = f(u,v,\varepsilon) \\ \dot{v} = \varepsilon g(u,v,\varepsilon) \end{cases}$, $0 < \varepsilon \ll 1$. If $\tau = \varepsilon t$, then $\begin{cases} \varepsilon u' = f(u,v,\varepsilon) \\ v' = g(u,v,\varepsilon) \end{cases}$ t: fast time; τ: slow time ($\varepsilon \neq 0$ 时等价) ↓ Defined in a neighborhood of $\mathcal{M}_0$

Reduced Problem: slow times $\varepsilon = 0$. **Layer Problem**: fast time $\varepsilon = 0$. **Critical Manifold**: $\mathcal{M}_0 := \{(u,v) : f(u,v,0) = 0\}$ ps: *Reduced problem* defined only on $\mathcal{M}_0$, *Layer problem* defined off $\mathcal{M}_0$

Normal Hyperbolicity: $\mathcal{M}_0$ is normally hyperbolic at $(u,v) \in \mathcal{M}_0$ if: λ_i (特征值) of Jacobian $J|_{(u,v,0)} = \frac{\partial f}{\partial u}|_{(u,v,0)}$ are $\Re(\lambda_i) \neq 0$. → **(Un-)Stable**: $\Re(\lambda_i) < 0_{(>0)} \Leftrightarrow \frac{\partial f}{\partial u} < 0_{(>0)}$ (1-dim)

Outer Solution: sol for slow time at $\varepsilon = 0$. (reduced problem) **Inner Solution**: sol for fast time at $\varepsilon = 0$. (layer problem) ↑ attracting(stable) / repelling(unstable)

Describe Reduced|Layer Problem: [1] 如果能解具体表达式就解; 不好解情况下给出 u', v' 仅关于 u, v 的表达式; [2] 如果可以画一个图; [3] 分析 attracting/repelling (stable/unstable); [4] 分析 $\tau, t \to \infty$ 时的行为

Fenichel's Theorem: If $\mathcal{M}_0 \subset \{f(u,v,0) = 0\}$ is compact (closed & bounded) and normally hyperbolic. Suppose f, g are *smooth*. Then for $\varepsilon > 0$ (sufficiently small), $\exists$ *Manifold* $\mathcal{M}_\varepsilon$ s.t. $\mathcal{O}(\varepsilon)$ close and diffeomorphic to $\mathcal{M}_0$. (i.e. $\mathcal{M}_\varepsilon$ is locally invariant under the flow of the full system (上面左边的 system)). ps: M_ε called **slow manifold**.

Find $\mathcal{M}_\varepsilon$: If $\mathcal{M}_0 = \{(u,v) : u = p_0(v)\}$, then $\mathcal{M}_\varepsilon = \{(u,v) : u = p_\varepsilon(v)\}$ 可通过算 u, v 的 exact sol, 寻找能使含 t 项消失但保留 τ 的解 (or: 考虑快变量 $t \to \infty$), 对比 $\mathcal{M}_0$ 得到 $\mathcal{M}_\varepsilon$

Invariance of Manifold: For slow $M_\varepsilon | M_0$ → manifold 中的等式关系 $u = p(v)$ 两边对 $\tau|t$ 求导 → 代入 slow|fast 对应的 u', v'; 代入 $M_\varepsilon | M_0$ 中的等式关系 → 判断是否 LHS=RHS. ↓ **Infective Period**: $= \frac{1}{\alpha}$

SIR: S: # people who are susceptible. (易感人群) I: # people who are infected. (感染者) R: # people who have been infected and recovered. (no longer susceptible). (康复者) **Total Pop**: $N = S + I + R$

Model: $S' = -\beta SI$, $I' = \beta SI - \alpha I$, $R' = \alpha I$, $S, I \geq 0$. **Assumptions:** [1] Mass Action incidence: βN transmit infection/unit time; [2] αI recover/unit time; [3] No im/emigration, births, deaths. [4] No deaths from disease. → 新感人数 $= \beta SI$ rate.

临界值: if $S > \frac{\alpha}{\beta}$, $I \uparrow$ 疾病爆发; If $S < \frac{\alpha}{\beta}$, $I \downarrow$ 疾病消退. ⇒ **Basic Reproduction Number**: (A threshold quantity) $R_0 = \frac{\beta S_0}{\alpha}$ or $\frac{\beta N}{\alpha}$, [1] $R_0 > 1$ epidemic occurs; [2] $R_0 < 1$ disease dies out.

$\cdot\ I_\infty = \lim_{t\to\infty} I(t) = 0$; $S_\infty = (S + I)_\infty > 0$; $N = S_0 + I_0$ **Final Size Relation**: $\ln(\frac{S_0}{S_\infty}) = \frac{\beta}{\alpha}[N - S_\infty] = R_0[1 - \frac{S_\infty}{N}]$ As $(S+I)' = -\alpha I \to N - S_\infty = \alpha\int_0^\infty I dt$; As $\frac{S'}{S} = -\beta I \to \ln(\frac{S_0}{S_\infty}) = \beta\int_0^\infty I dt$

Attack Rate/Ratio: $N - S_\infty$ (# people ever infected | 总感染人数 | Final size of epidemic) **Orbits Sol**: $I(t) + S(t) - \frac{\alpha}{\beta}\log(S(t)) = N - \frac{\alpha}{\beta}\log(S_0)$ **Peak**: $I_{\max}$ when $S = \frac{\alpha}{\beta} \leftrightarrow I' = 0$ (代入前面得精确 $I_{\max}$)

6 Reaction Diffusion

Diffusion: Net movement from high to low concentration through random, small-scale motion. **Fickian Diffusion:** $J = -D\frac{\partial c}{\partial x}$ (J: flux; D: diffusivity; c: concentration; $-$: from high to low)

Conservation Realtion: $\frac{\partial}{\partial t}\int_{x_0}^{x_1} c(x,t)dx = J(x_0,t) - J(x_1,t)$ $x_1 = x_0 + \Delta x$ When $\Delta x \to 0 \Rightarrow \frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} = \frac{\partial(D\frac{\partial c}{\partial x})}{\partial x}$ (If D const. Heat Equation)

Solve PDE: For $u(\mathbf{x},t)$, $\mathbf{x} = (x_1, \ldots, x_n)$; $u_t = D\Delta u$ with $u(\mathbf{x},0) = f(\mathbf{x})$ If $f(\mathbf{x}) = Q\delta(\mathbf{x})$, then **Fundamental Sol**: $u(\mathbf{x},t) = \phi(\mathbf{x},t) = \frac{Q}{(4\pi Dt)^{n/2}}e^{-\frac{|\mathbf{x}|^2}{4Dt}}$ Else: **General Sol**: $u(\mathbf{x},t) = \int_{\mathbb{R}^n}\phi(\mathbf{x}-\mathbf{y},t)f(\mathbf{y})d\mathbf{y}$

Moreover, for $u(\mathbf{x},t)$, $u_t = D\Delta u + a(t)u$ with $u(\mathbf{x},0) = f(\mathbf{x})$ ⇒ $u(\mathbf{x},t) = v(\mathbf{x},t)e^{\int_0^t a(s)ds}$ with $v(\mathbf{x},0) = f(\mathbf{x})$ satisfying $v_t = D\Delta v$. → Use above. **invades const. radial speed**: 算 $\frac{dr}{dt}$

Remark for Boundary: [1] 任何 Diffusion Model: 认为 $u(\mathbf{x},t)$ 是 **bounded** 的. (含 $\pm\infty$, 极坐标 $r \to \infty$, 得 IC) [2] 对 u, J (flux), 有 **continuous**. (得 BC, 尤其是分段函数) [3] 对 J (flux), 考虑由物理意义, 边界是否为 0

假设和默认: [1] 若要假设 $u(\mathbf{x},t)$ 的 IC, 考虑 $u(\mathbf{x},0) = b_0 > 0$ (均匀的分布) [2] **Stationary Sol**: For $u(\mathbf{x},t)$, Let $u_t = 0$ to find sol. [3] **Homogeneous Rest State**: For $u(\mathbf{x},t)$, Let $u_{\mathbf{x}} = 0$ and $u_t = 0$ to find sol.

Useful Tools: [1] 最后, 可通过两边对 u 对 x 积分得待定系数值 [2] **Linearise at** u^*: Let $u = u^* + \eta$, $|\eta| \ll 1$ func. [3] 对于求解 PDE, 考虑直接计算 or Seperation of var. [4] **Desingularise**: e.g. For $u(x,t)$, Let $\tau = ut$

7 Appendix

$y' + p(x)y = q(x)$: **Integrating Factor**: $I(x) = e^{\int p(x)dx}$ ⇒ $I(x)y = \int I(x)q(x)dx$ **Taylor Expansion:** $f(x,r)$ at (x_0,r_0): $f(x,r) = f(x_0,r_0) + (x-x_0)f_x|_{(x_0,r_0)} + (r-r_0)f_r|_{(x_0,r_0)} + \frac{1}{2}\cdots$

Binomial Series: $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots$ **Linerisation:** $f(x+\Delta x) = f(x) + f'(x)\Delta x + \cdots$ $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \cdots$ $\ln(a+x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \cdots$ $y(a) - y(b) = \int_b^a y'(x)dx$

Geometry Series: $\frac{1}{1-x} = 1 + x + x^2 + \cdots$, $|x| < 1$ **Hyperbolic Funcs:** $\sinh x = \frac{e^x - e^{-x}}{2}$ $\cosh x = \frac{e^x + e^{-x}}{2}$ $\tanh x = \frac{\sinh x}{\cosh x}$ $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$ $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \cdots$

Integral: $\operatorname{arctanh}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$ $\operatorname{arcsinh}(x) = \ln\left(x + \sqrt{x^2+1}\right)$ $\operatorname{arccosh}(x) = \ln\left(x + \sqrt{x^2-1}\right)$ $\int\frac{1}{\sqrt{a^2-x^2}}dx = \arcsin(\frac{x}{a})$ $\int\frac{1}{a^2+x^2}dx = \frac{1}{a}\arctan(\frac{x}{a})$

Useful Tools: $\Delta u = u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$ If u **radial symmetry**: $u = u(r,t)$ (i.e. $u_\theta = 0$) $dA = rdrd\theta$ **in polar coord.** and $dV = r^2\sin\phi\, drd\theta d\phi$ **in spherical coord.**

$ay'' + by' + cy = 0$: $y'' - \lambda y = 0$: [1] $\lambda > 0$: $y = C_1\cosh(\sqrt{\lambda}x) + C_2\sinh(\sqrt{\lambda}x) = Ae^{\sqrt{\lambda}x} + Be^{-\sqrt{\lambda}x}$; [2] $\lambda = 0$: $y = C_1 + C_2x$; [3] $\lambda < 0$: $y = C_1\cos(\sqrt{-\lambda}x) + C_2\sin(\sqrt{-\lambda}x)$

General: *characteristic eq.*: $ar^2 + br + c = 0$, root $r_{1,2} = \mu \pm \nu i$. [1] $\Delta > 0$: $y = C_1e^{r_1x} + C_2e^{r_2x}$; [2] $\Delta = 0$: $y = e^{\mu x}(C_1 + C_2x)$; [3] $\Delta < 0$: $y = e^{\mu x}(C_1\cos(\nu x) + C_2\sin(\nu x))$

Seperation of var.: [1] 假设 $u(x,t) = X(x)T(t)$ → 带入 PDE [2] 解出类似 $\frac{T'}{DT} = \frac{X''}{X} = -\lambda$ → 得到两个 ODE [3] 解较为简单的 ODE, 代入 BC 解 λ 和 sol. (注意看是否只要 periodic; 极坐标有 $r \to \infty$ bounded, $\theta = \theta + 2\pi$ 等条件), [4] 解另外一个. [5] 叠加所有 λ_n 的 sol 得到 PDE 的 general sol $u = \sum c_nX_n(x)T_n(t)$ [6] 用 IC 解出 c_n. Remark: [1] 前期可以用正比化简; [2] λ 为让 sin/cos 解为正好计算.